EBU5376 Digital Signal Processing

LAB 1

Before you start, please read the following information carefully:

General Instructions:

- The .zip file you downloaded from QMPlus should contain the following documents:
  - This lab sheet (needs to be completed)
  - A WAV audio file that you need for both Question 1 and Question 2
- Type your answers in the textboxes provided. Figures need to be copied into the textboxes. If you need more space, enlarge the textbox or add a new box.
- Answers outside the textboxes will NOT be marked.
- If you require any technical assistance, please attend the lab session and ask the on-site TA for help.

How to submit:

- Submit one .zip file to QMPlus. The .zip file should contain:
  1. This LAB SHEET converted to a PDF document
  2. The MATLAB script containing all the work you did

Additional grading criteria:

- Your answers in this sheet must include sufficient explanations/derivations. Marks will be deducted if no explanation is given to support the answer.
- Your MATLAB script will be checked from the following three aspects:
  - Whether it can **REPRODUCE ALL THE FIGURES** in this sheet
  - Whether you have added appropriate **COMMENTS** to explain the codes
  - Plagiarism
- If the MATLAB script is not submitted, your grade will be capped to 60%.
- Email submission NOT accepted.
- Do NOT plagiarize. Both the MATLAB script and this sheet will be checked for plagiarism.

Question 1 – Basic Signal Processing **[25 marks]**

In this question, you will practice how to load audio files, process and visualize audio data, and explore the characteristics of sound using MATLAB.

In Lab 1 materials, you are given a WAV audio file that consists of a sequence of pure music notes.

Task 1: Use the command 'audioread( )' to read the audio in MATLAB, then answer the following questions:

1. What is the sampling rate of this signal?
2. How many samples are there in this signal?
3. What is the length (in seconds) of this signal?

(Optional: You can use the command 'soundsc( )' to play the audio in MATLAB.)

[4 marks]

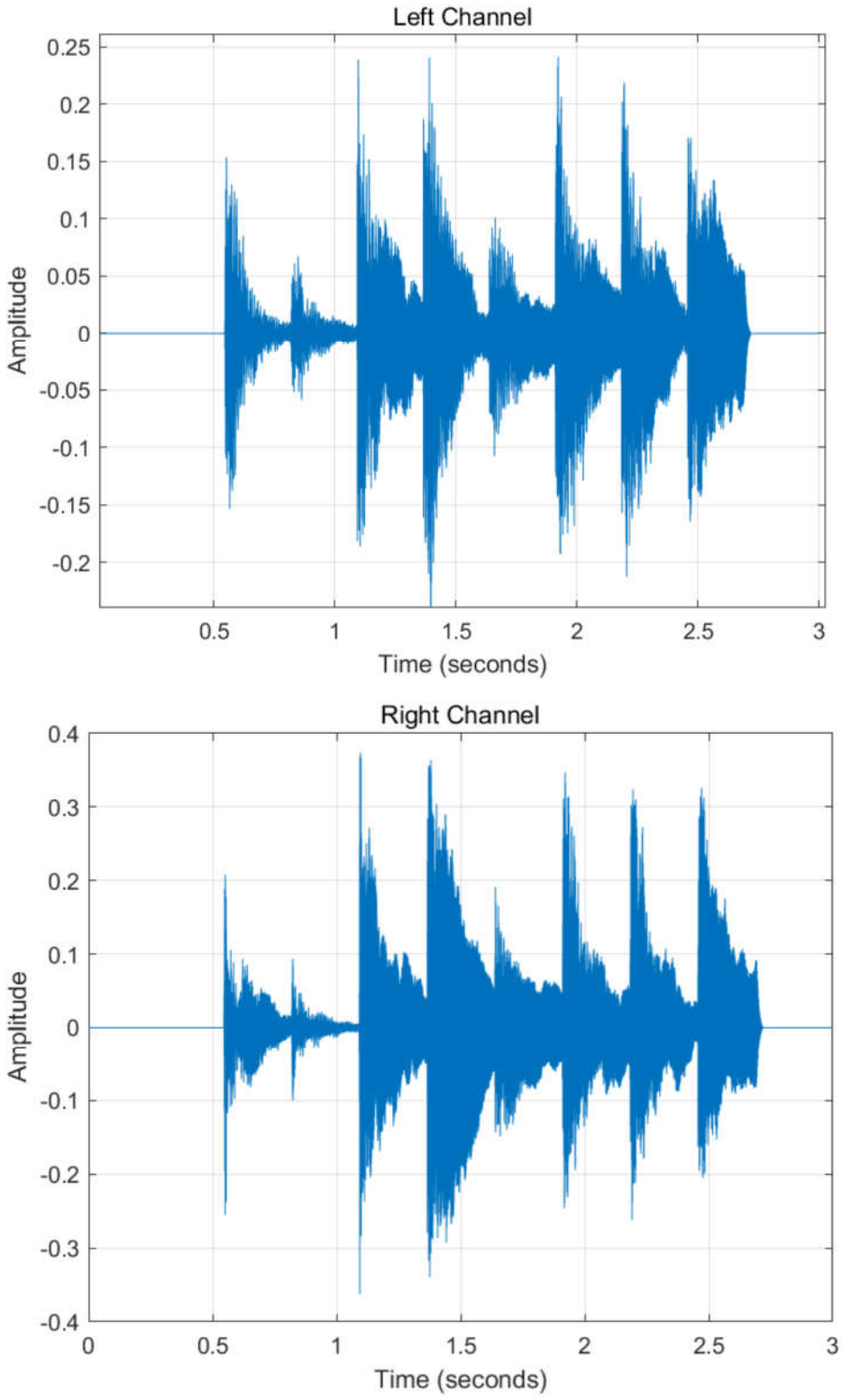
Q1 Task 1.1: Sampling Rate (Fs) = 48000 Hz
Q1 Task 1.2: Number of Samples = 144000
Q1 Task 1.3: Signal Duration = 3.00 seconds

Task 2: The provided audio file is stereo, with two channels (left channel and right channel) of audio data. Hence, the audio data should contain two dimensions that represent the samples over time and the stereo channels. On **two separate figures**, plot the left channel and the right channel of this audio signal, with the following requirements:

1. The horizontal axis is labelled in time (seconds).
2. Using titles or legends, clearly label whether the plot is the left channel or right channel.

[6 marks]

[Paste the figures here]



Task 3: The audio signal consists of several pure music notes. By inspecting the plot, how many music notes are there in this audio?

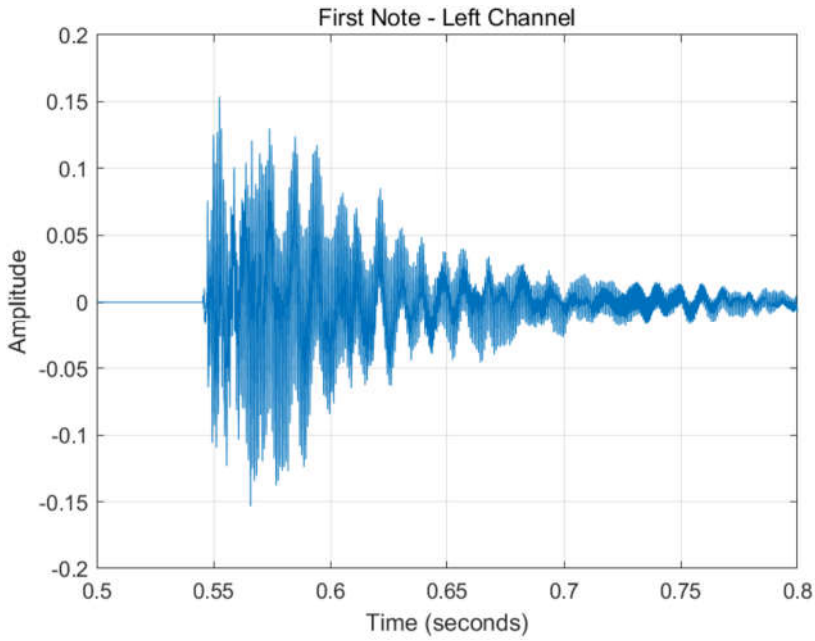
[1 mark]

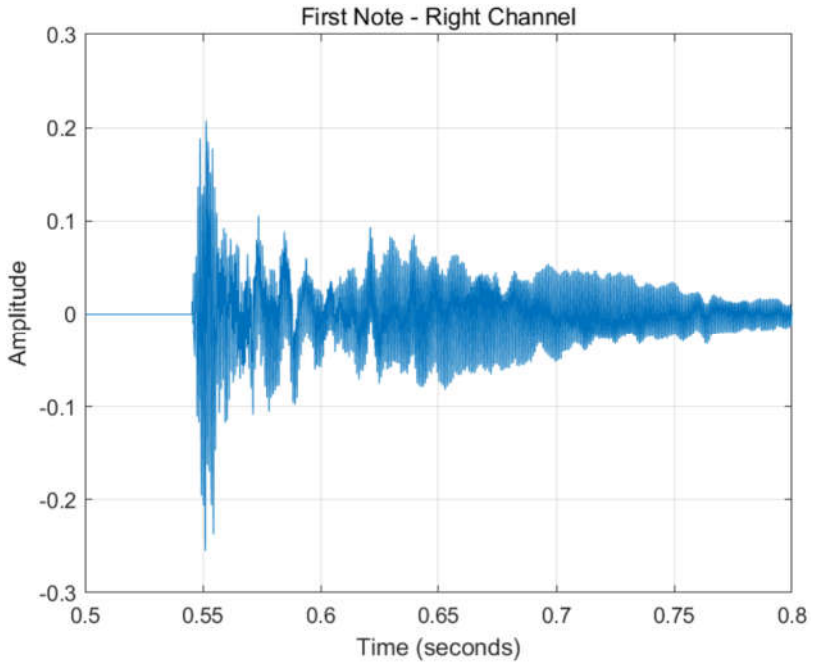
[Type your answers here]

8 notes

Task 4: We would like to take a closer look at the **first music note**. On separate figures, plot the left and right channels for the segment where the first note is played. Clearly indicate whether the plot shows the left or right channel.

[6 marks]

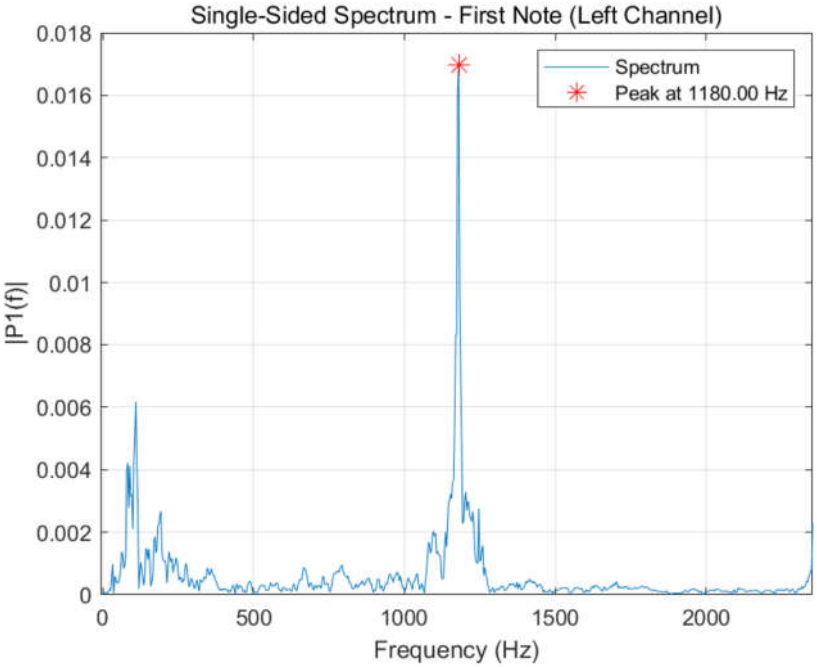


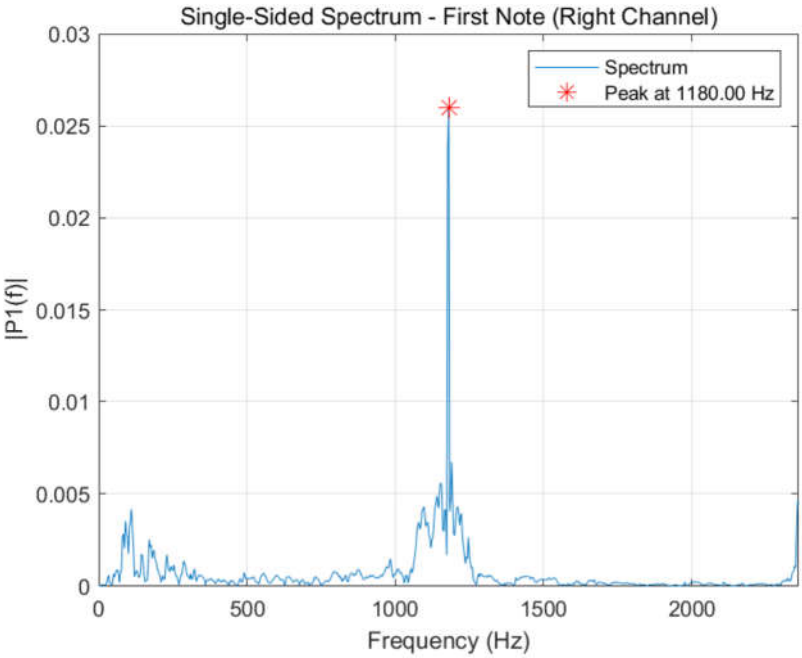


Task 5: Using the Fourier transform, estimate the frequency (in Hz) of the **first music note** in the given audio signal. Carefully explain the steps taken to obtain the answer and include figures where relevant.

[8 marks]

[Type your answers here]





Estimated frequency of the first note: 1180.00 Hz

1. Extracted the time-domain signal for the first note (e.g., left channel from 0.500000 s to 0.800000 s).
2. Applied the Fast Fourier Transform (FFT) using the `fft` function.
3. Calculated the corresponding frequency axis based on the sampling rate (Fs) and the number of points in the FFT (N).
4. Computed the single-sided amplitude spectrum to represent positive frequencies.
5. Found the frequency corresponding to the maximum peak in the amplitude spectrum.
6. Plotted the spectrum and marked the peak frequency.

Question 2 – Advanced Frequency Analysis **[25 marks]**

In this question, you will extend your analysis from Question 1 to estimate the frequencies of all music notes in the audio signal. You will also reflect on the process and discuss key observations and factors affecting the accuracy of your results.

Task 1: Use Fourier transform to estimate the frequency of all music notes in the given audio. Directly write down your results, you don't need to show your work here.

[8 marks]

Note 1 (0.550000 s - 0.750000 s): 1180.00 Hz

Note 2 (0.800000 s - 1.000000 s): 1050.00 Hz

Note 3 (1.100000 s - 1.300000 s): 835.00 Hz

Note 4 (1.400000 s - 1.600000 s): 790.00 Hz

Note 5 (1.650000 s - 1.850000 s): 700.00 Hz

Note 6 (1.900000 s - 2.100000 s): 625.00 Hz

Note 7 (2.200000 s - 2.400000 s): 590.00 Hz

Note 8 (2.450000 s - 2.650000 s): 525.00 Hz

Task 2: Briefly describe how the **time window of the notes** (i.e., **start and end time of each note**) were decided in your answers.

[2 marks]

Explain that you visually identified segments corresponding to individual notes, looking for:

- Clear beginnings and ends of sustained oscillations.
- Periods of relative silence or rapid decay between notes.

Note 1 (0.550000 s - 0.750000 s)

Note 2 (0.800000 s - 1.000000 s)

Note 3 (1.100000 s - 1.300000 s)

Note 4 (1.400000 s - 1.600000 s)

Note 5 (1.650000 s - 1.850000 s)

Note 6 (1.900000 s - 2.100000 s)

Note 7 (2.200000 s - 2.400000 s)

Note 8 (2.450000 s - 2.650000 s)

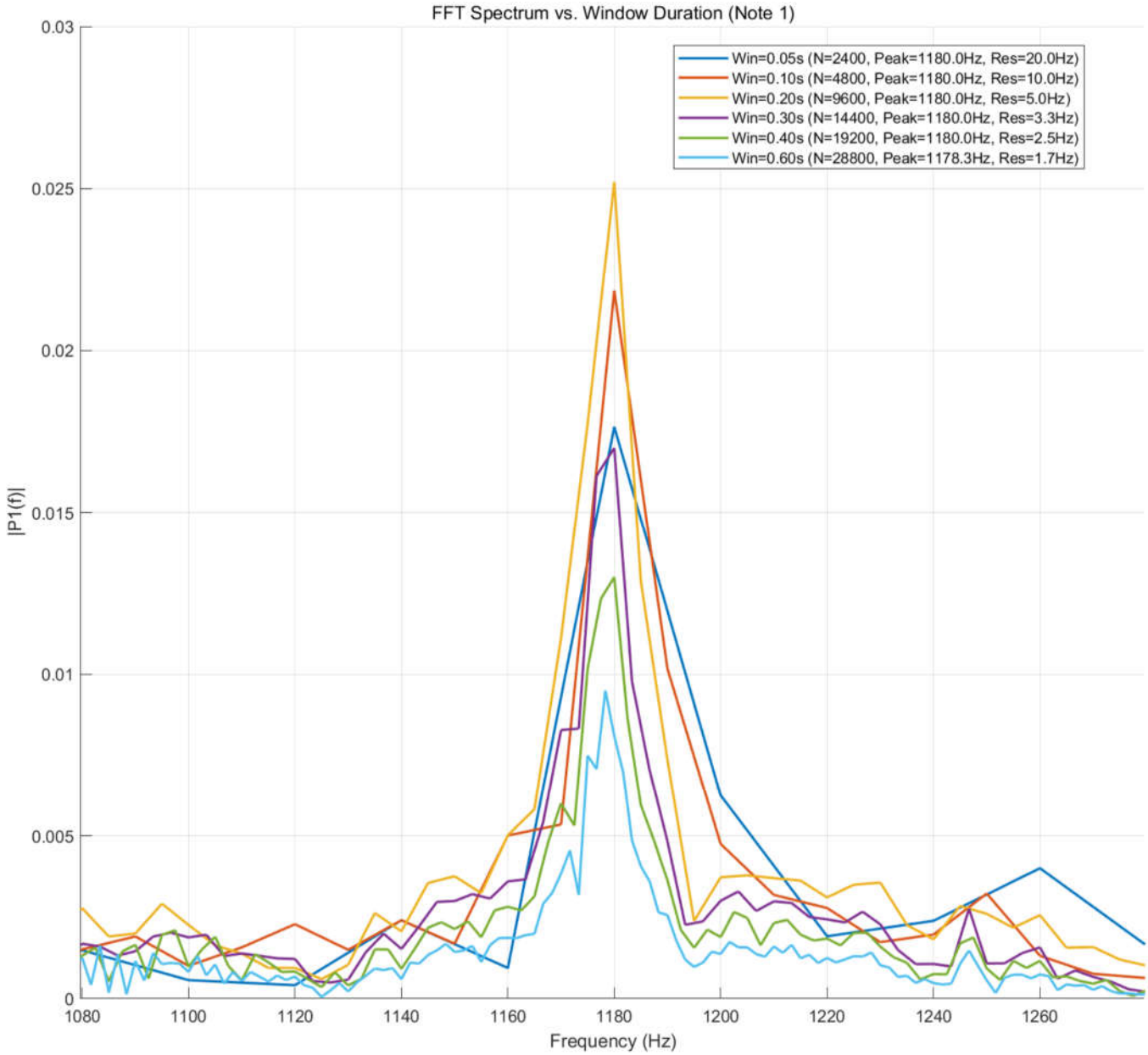
Task 3: For this particular audio signal, investigate how the duration of the time window used to isolate notes impacts the Fourier spectrum and the accuracy of estimated frequencies. Think about

1. How does the **peak frequency** change with window length? Is it stable or shifting?
2. What happens to the **frequency resolution** (width of peaks) in the spectrum as the window size changes?
3. How to set the window length for ensuring a reliable frequency estimate?

You need to show detailed analysis and figures to support your conclusion. Make sure your answers are clear and well-structured.

[15 marks]

[Type your answers here]



1. **Peak Frequency Stability:**

- Observing the legend, the estimated peak frequency remains remarkably stable at **1180.0 Hz** for window durations ranging from 0.05s up to 0.40s.
- There is only a very slight shift to **1178.3 Hz** for the longest window duration tested (0.60s).
- Visually, the peaks of all spectrum plots (except perhaps the light blue 0.60s one) align very closely on the frequency axis around 1180 Hz.
- **Conclusion:** For this particular note, the estimated peak frequency is highly stable across a wide range of window lengths, suggesting 1180 Hz is a reliable estimate of the fundamental frequency. The slight shift at the longest duration might indicate the window is starting to include less representative parts of the signal (e.g., decay) or is simply reflecting the discrete nature of the FFT bins at higher resolution.

2. **Frequency Resolution (Peak Width):**

- The frequency resolution is defined as Fs/N*Fs*/*N*, where Fs*Fs* is the sampling rate and N*N* is the number of samples in the window (which increases with window duration). A smaller Fs/N*Fs*/*N* value indicates better resolution.
- The legend clearly shows that as the window duration increases, the frequency resolution (Res) improves (decreases): from 20.0 Hz for the 0.05s window down to 1.7 Hz for the 0.60s window.
- This is visually confirmed in the plot:
- The spectrum for the shortest window (0.05s, dark blue) has a very broad, poorly defined peak.
- As the window duration increases (e.g., 0.10s orange, 0.20s yellow, 0.30s purple, 0.40s green), the peaks become progressively **narrower and sharper**.
- The spectrum for the longest window (0.60s, light blue) exhibits the sharpest, most well-defined peak.
- **Conclusion:** Increasing the window duration significantly improves the frequency resolution, resulting in narrower peaks in the spectrum. This makes it easier to pinpoint the exact peak frequency.

3. **Setting Reliable Window Length:**

- **Trade-offs:** There's a trade-off between window length, resolution, and potentially including unwanted signal portions.
- *Too short:* While the 0.05s window yielded the correct peak frequency *here*, its very poor resolution (20 Hz) and broad peak make the estimate less reliable in general. It might easily latch onto noise or be unstable for other signals.
- *Too long:* The 0.60s window, despite having the best resolution (1.7 Hz), showed a slight frequency shift. This might indicate it's capturing part of the note's decay or extending slightly beyond the stable part of the note, potentially slightly skewing the main frequency component. It could also risk including silence or adjacent sounds if notes were closer together.
- *Appropriate Range:* Windows from 0.20s to 0.40s provide a good balance for this note. They yield the stable peak frequency of 1180.0 Hz and offer significantly improved resolution (5.0 Hz down to 2.5 Hz) compared to shorter windows, resulting in reasonably sharp peaks.

Final Evaluation (optional): Copy and paste the following codes to MATLAB. Replace the first two lines with your results. Then, you will be able to hear the music sequence based on your estimation. You may compare it to the given audio.

```
frequencies = [f1, f2, f3, f4, f5, f6, f7, f8]; % replace f1-f8 with your estimations in Q2 task 1
fs = sample_rate;        % set this to the sampling rate of the audio, calculated in Q1 task 1

note_duration = 0.3;
rest_duration = 0.05;
fade_duration = 0.2;

audio_signal = [];
for i = 1:length(frequencies)

   t = 0:1/fs:note_duration-1/fs;
   note = sin(2*pi*frequencies(i)*t);

   fade_samples = round(fade_duration*fs);
   fade_window = ones(1, length(note));
   fade_window(end-fade_samples+1:end) = linspace(1, 0, fade_samples);
   note = note .* fade_window;

   audio_signal = [audio_signal, note];

   if i < length(frequencies)
      audio_signal = [audio_signal, zeros(1, round(rest_duration*fs))];
   end
end

audio_signal = audio_signal / max(abs(audio_signal));

% Play the sequence
sound(audio_signal*0.2, fs);
```